

backend/core/finetuner.py Documentation (Roman Urdu)

Yeh file application ke andar mojud ek mukammal **AI Training Engine** hai. Iska maqsad yeh hai ke developers ya users apne local models (jaise Llama-3 ya Mistral) ko apni personal chat history par fine-tune (tarbiyat) kar sakein.

File Ka Maqsad (Purpose)

UI m Training Start ka button dabaate hi yeh script kaam me ati hai. Yeh automatically HuggingFace libraries (`transformers`, `peft`, `trl`) ka setup karti hai. Agar PC mein NVIDIA Graphic Card (GPU) ho toh yeh model ko 4-bit (QLoRA) pe train karta hai, aur agar GPU chota hai toh fallback k tor pr purely CPU par load karta hai.

Code Blocks aur Classes ki Tafseel

1. Dictionaries (`OLLAMA_TO_HF_MAP` / `MODEL_VRAM_4BIT_GB`)

- **Kya karti hain?:** Yeh uper define ki hui maps hain jo Ollama ke local models (`llama3.2:1b`) ko unke asli HuggingFace ID (`meta-llama/Llama-3.2-1B-Instruct`) se jodti hain taake internet se base weights download k sakte ja sakein. Dosri dictionaries mein estimated memory likhi hui hai (jaise 7B ko train honay k liae 10GB vram chahiye).

2. `class FineTuner`

- Yeh class pure training procedure ko lead (manage) karti hai.
- `get_training_capabilities()`: System ki RAM aur VRAM parh ke user ko UI pe alert deti hai k training "GPU pe hogi" ya "CPU pe shift ho ri ha".
- `_run_training()` / `async start_training()`: Yeh main async loop hai. UI se raw data jsonl format me save krke, yeh completely aik naya background python process start karwati hai.
- `_write_training_script()`: Yeh ek buhat tricky aur smart code hissa hai. FastAPI server RAM heavy hojata hai isliye `FineTuner` aik completely alag script (`train_script.py`) string generate kar ke physical folder mein write karta hai aur background cmd m run (`subprocess`) kr deta hai wth proper progress events.

- `_apply_progress_line()`: Background me chalne wala process `print("PROGRESS:...")` emit karta hai, jisy parh k UI m percentage bar bharta hai (Loss aur Steps map kar k).
- `_register_with_ollama()`: Jab training finish hojati hai toh `model.merge_and_unload()` se LoRA file ko base model m mila (merge) diya jata hai, aur `ollama create` ki command fire kar k Ollama interface mein directly add kr dia jata hai (e.g. `MySoloBot-v1`).